



Slicing the Serpent: Markov Chain Monte Carlo and the Hidden Topography of the Recovery Ice Stream

Noelio Perez, Niya Shao, Dr. Emma MacKie

University of Florida



Overview

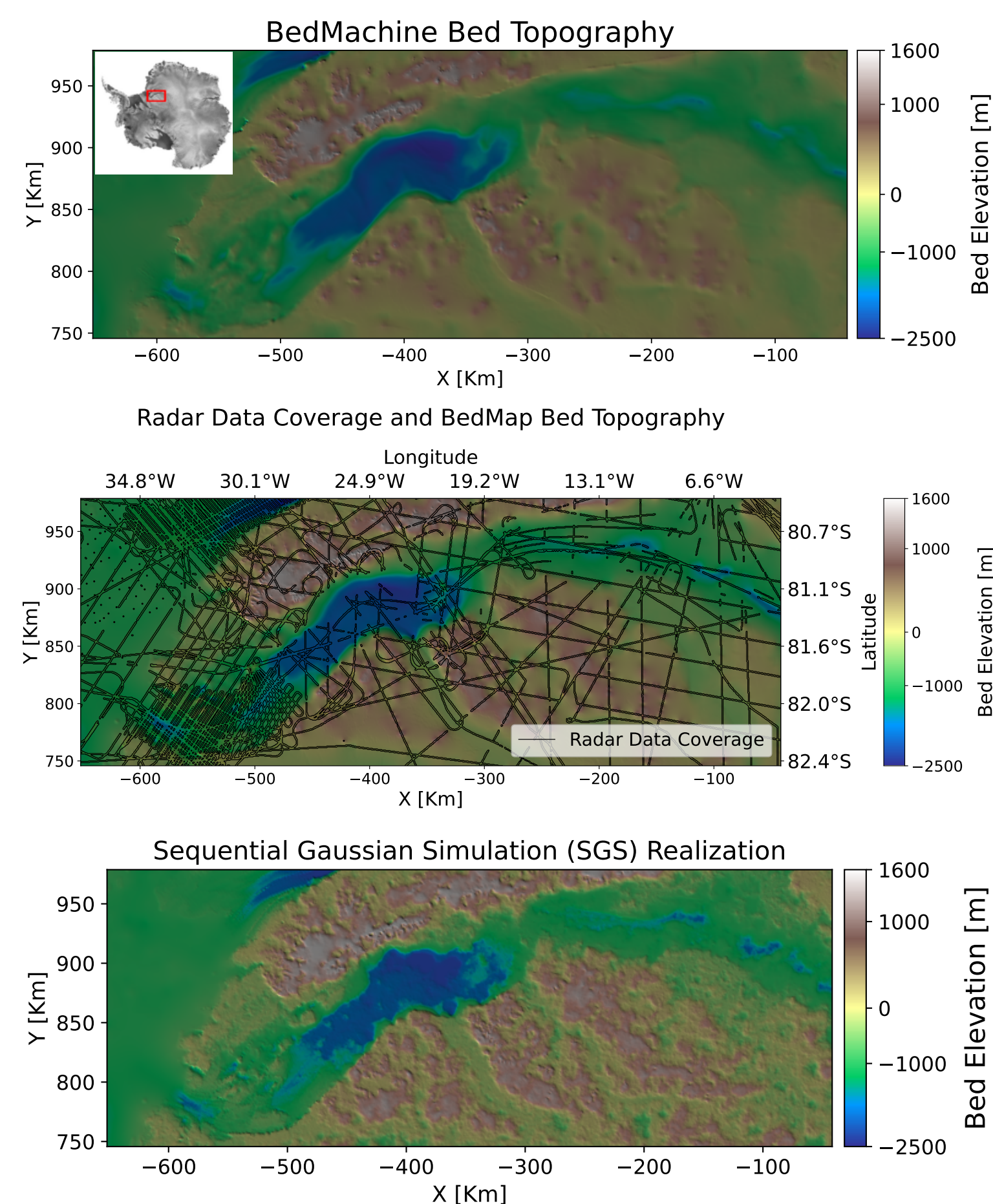
The Recovery Ice Stream is one of Antarctica's longest and fastest glaciers, sliding **35 billion tons** of ice into the ocean each year.

Due to its vital importance to the Antarctic ecosystem and its contributions to climate change, a powerful method is needed to quantify its bed topography.

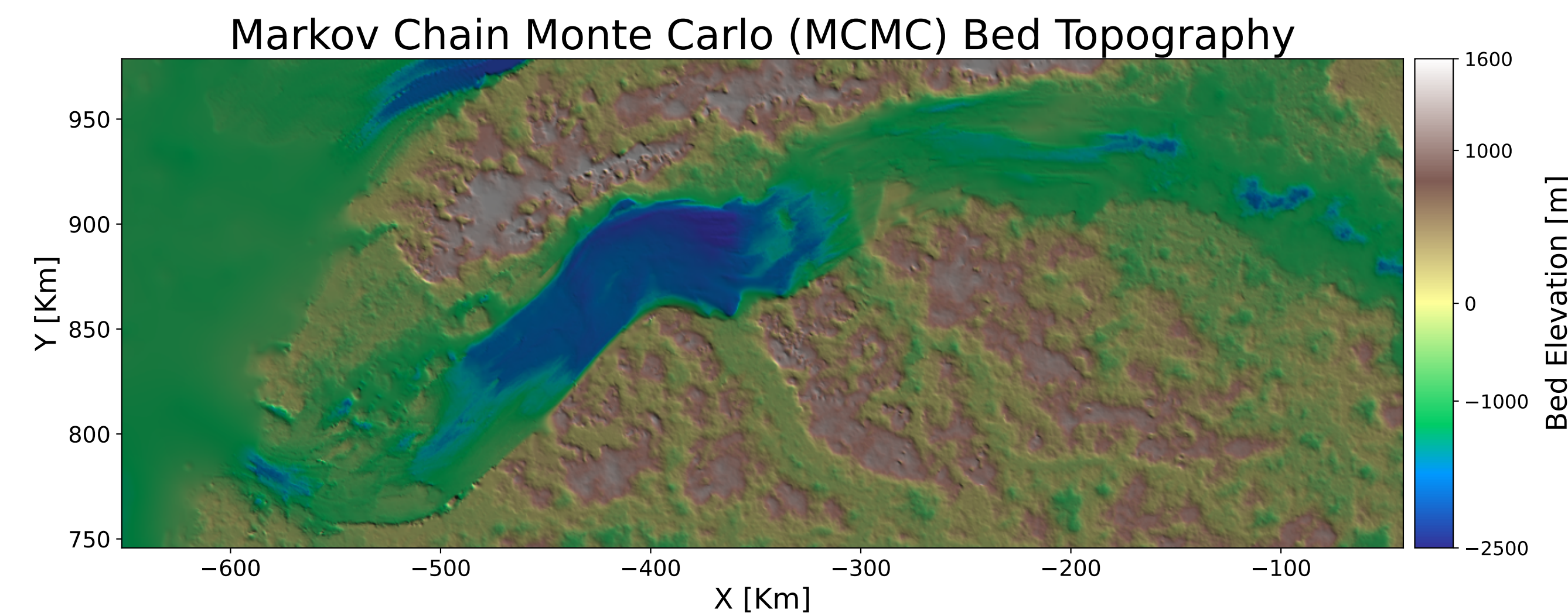
With this new method, we can prioritize both good topographical roughness and low mass conservation residuals.

Methods

We used a **Sequential Gaussian Simulation (SGS)** to generate a realistically detailed topographic map, and then improved its ice flux residuals with a **Markov Chain Monte Carlo (MCMC)** algorithm.



MCMC Bed Topography



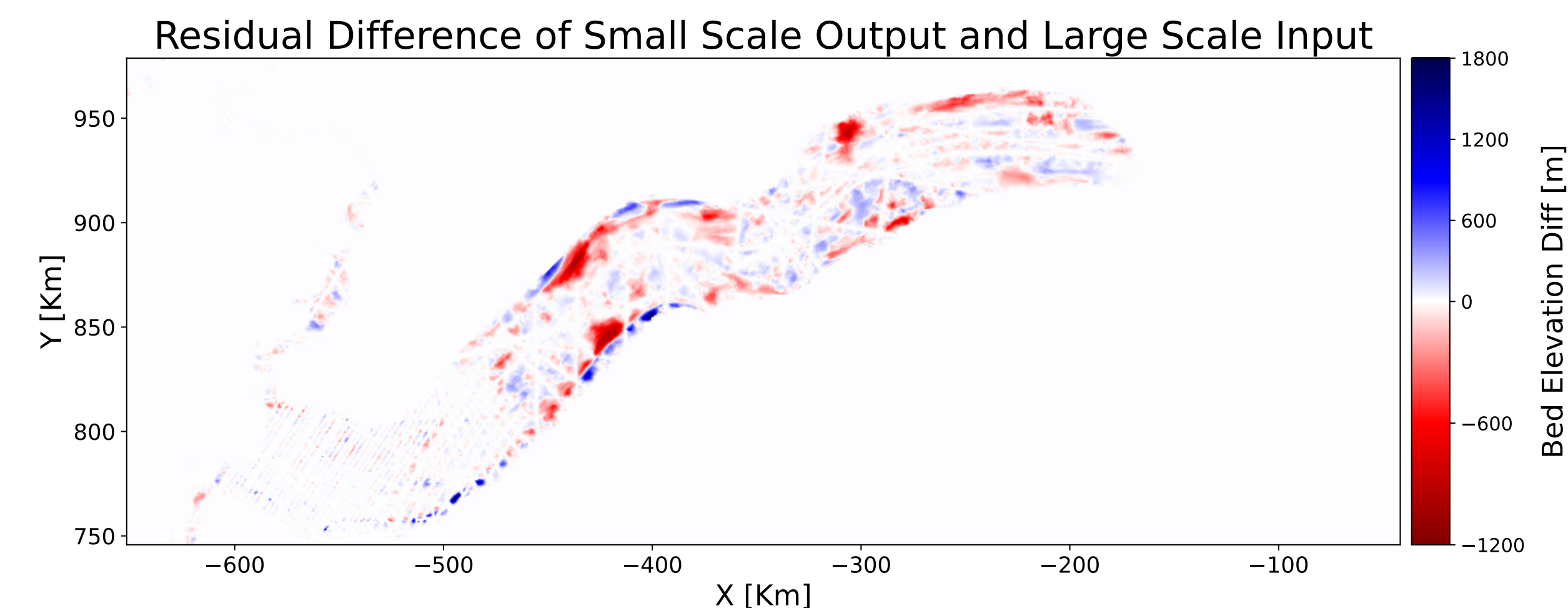
Results

After a large-scale chain run of **7 million iterations** and a small-scale chain run of **100K iterations**, the MCMC algorithm reduced the sum of squared mass flux residuals loss from **2.632e6** to **1.540e6**.

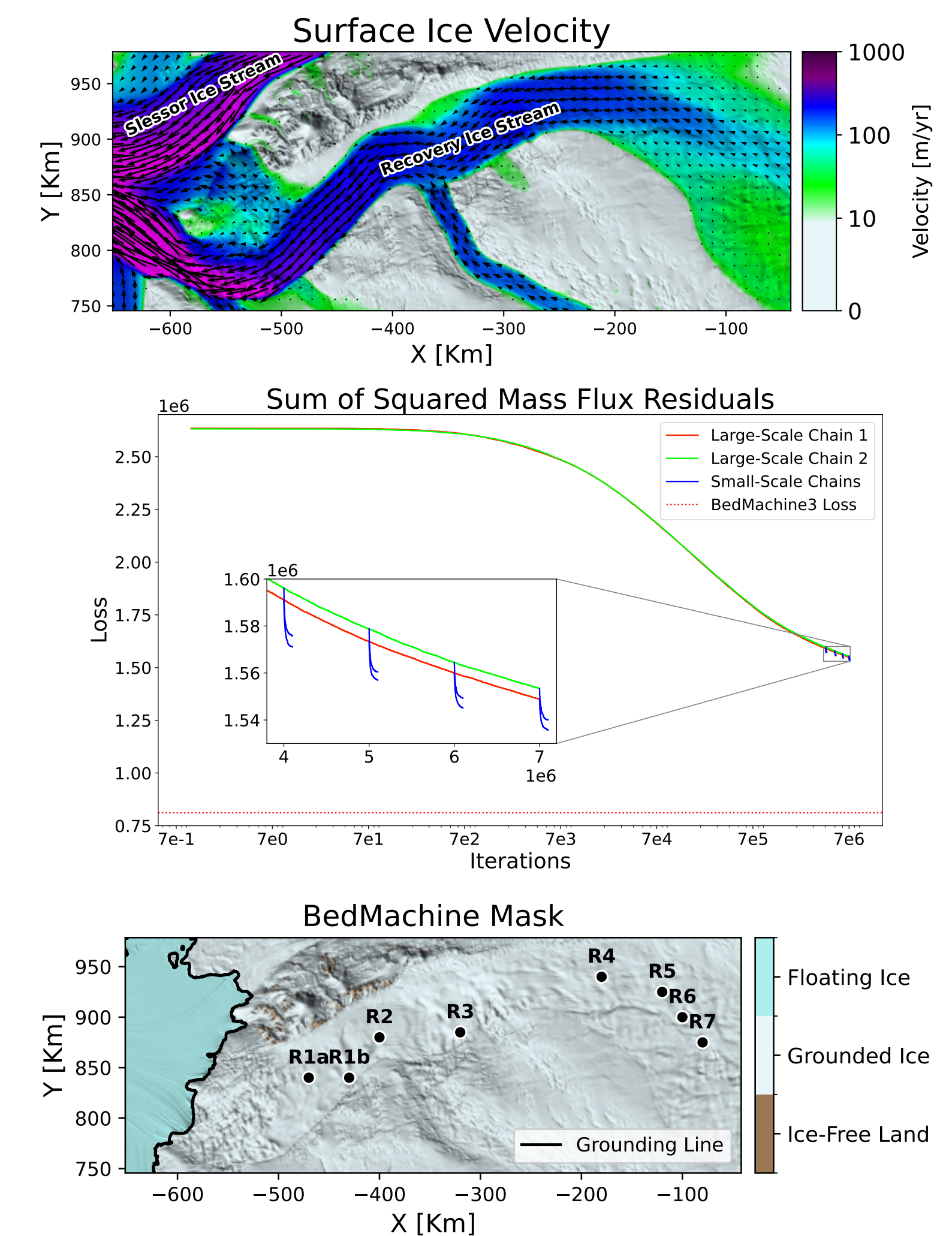
It creates a topographical map with similar roughness to SGS, but with a considerably lower loss.

Note: BedMachine3 loss is 8.115e5. More iterations (30M+) are needed for full convergence.

Bed Difference



Velocity, Loss, & Masks



GitHub Repository



NoelioPerez09/Recovery-MCMC